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Expandable spinal implant system and method of using same

Abstract

An expandable spinal implant includes a distal projection extending from only one side of the implant, ending in an anterior tip, the anterior portion and anterior tip defining an elongated distal end hook, which is wider than the proximal end. The distal end hook rotates around the spinal cord, aligning the implant with a desired pathway, then inserts into place in the disc space between the vertebrae. The elongated widened distal end hook provides a TLIF approach, distributes loads, provides anterior rim engagement, and creates lordosis.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. application Ser. No. 17/186,862, filed Feb. 26, 2021; which is continuation of U.S. application Ser. No. 16/295,714, filed Mar. 7, 2019, now U.S. Pat. No. 11,065,130; which is a divisional of U.S. application Ser. No. 15/340,770, filed Nov. 1, 2016, now U.S. Pat. No. 10,238,503; all of which are incorporated by reference herein.

TECHNICAL FIELD

(1) The present disclosure generally relates to medical devices for the treatment of musculoskeletal disorders, and more particularly to a surgical system that includes an expandable spinal implant, systems for implanting an expandable spinal implant, and a method for treating a spine.

BACKGROUND

(2) Spinal disorders such as degenerative disc disease, disc herniation, osteoporosis, spondylolisthesis, stenosis, scoliosis and other curvature abnormalities, kyphosis, tumor, and fracture may result from factors including trauma, disease and degenerative conditions caused by injury and aging. Spinal disorders typically result in symptoms including pain, nerve damage, and partial or complete loss of mobility.

(3) Non-surgical treatments, such as medication, rehabilitation and exercise can be effective, however, may fail to relieve the symptoms associated with these disorders. Surgical treatment of these spinal disorders includes fusion, fixation, correction, discectomy, laminectomy and implantable prosthetics. As part of these surgical treatments, spinal constructs, such as, for example, bone fasteners, spinal rods and interbody devices can be used to provide stability to a treated region. For example, during surgical treatment, interbody devices may be introduced to a space between adjacent vertebral bodies (the interbody space) to properly space the vertebral bodies and provide a receptacle for bone growth promoting materials.

(4) More recently, interbody devices have been introduced that provide additional capability beyond static spacing of the vertebral bodies. For example, some devices have expansion capability such that the implant may be introduced to the interbody space in a collapsed state and then expanded to produce additional spacing and, in some cases, introduce or restore curvature to the

spine by expanding selectively on only one end or portion of the implant. However, many existing expandable interbody designs utilize internal mechanisms that may inhibit the introduction of bone growth promoting material into the interbody implant by a surgeon after the implant is expanded. (5) An additional problem exists related to subsidence of spinal surfaces due to existing interbody devices having inadequately-sized load-bearing surfaces. In the case of expandable devices, the loads on the load-bearing surfaces, including loads generated during expansion of the implant, are often significant. An expandable implant with relatively large surface areas is needed to bear the loads, including the loads generated during implant expansion, in an attempt to avoid a need for follow-on surgery due to subsidence of spinal surfaces.

(6) The present invention seeks to address this and other shortcomings in the existing art.

SUMMARY

(7) In one embodiment, an expandable spinal implant is provided deployable between a collapsed position and an expanded position in a disc space between upper and lower vertebral bodies. The implant includes a frame comprising a proximal wall, lateral walls, and a distal wall, wherein the proximal wall defines a proximal aperture and the distal wall defines a distal aperture. The implant also includes a plug movably disposed in the distal aperture of the frame and first and second endplates operably engaged with the frame and configured to expand outward from the frame when the plug is moved in a distal direction relative to the frame, each of the first and second endplates including respective first and second lateral walls and respective distal ends.

(8) In one alternative embodiment a system is provided including an expandable spinal implant and an insertion instrument. The insertion instrument comprises a cannulated outer shaft and a driver shaft removably and rotatably disposed within the cannulated outer shaft. The expandable spinal implant comprises a frame with a proximal wall and a distal wall, wherein the proximal wall defines a proximal aperture and the distal wall defines a distal aperture. The proximal wall of the frame is configured to receive a distal end of the cannulated outer shaft for manipulating the expandable spinal implant. The expandable spinal implant also comprises a movable plug disposed in the distal aperture of the frame, wherein the plug comprises an interface configured to be operably engaged by a distal end of the driver shaft to move the plug relative to the frame. The expandable spinal implant also comprises first and second endplates engaged with the frame and configured to move relative to the frame when the plug is moved by the driver shaft of the insertion instrument. The driver shaft is also configured to be removable from the cannulated outer shaft of the insertion instrument such that after the plug has been moved distally relative to the frame, a bone growth promoting material may be introduced into the frame through the cannulated outer shaft of the insertion instrument.

(9) In one alternative embodiment, each of the first and second endplates include an anterior portion projecting from the first and second lateral walls at the respective distal ends thereon. The projecting anterior portions increase the surface area of the respective endplates, and end in respective anterior tips. When the implant is in the collapsed position, the anterior portions and the anterior tips define a distal end hook portion. In addition, when the implant is in the collapsed position, the first endplate distal end and the second endplate distal end define a distal end beveled portion. The distal end hook portion has a greater width than the proximal end of the implant. The distal end hook portion is configured, upon insertion of the implant into the disc space, to hook at least partially around the vertebral foramen, thereby avoiding interference with the spinal cord, and other neural elements located within the vertebral foramen. The distal end hook portion is further configured, upon insertion into the disc space, to rotate to a TLIF/transverse pathway. The implant is now positioned to be inserted to the anterior apophyseal rims of each of the upper and lower vertebral bodies. Upon expansion of the implant, the distal hook end portion and the distal beveled end portion engage substantially the entire anterior apophyseal rim of each of the upper and lower vertebrae. Rows of teeth provided on the first endplate and on the second endplate prevent inadvertent backing out of the implant from the anterior apophyseal rims. The additional surface

area and the additional width of at least the distal end hook portion enables the endplates to bear significantly greater loads applied to the implant by the upper and lower vertebrae. The distal end hook configuration enables a single implant to be positioned across the midline of the disc space, a single implant to be positioned spaced laterally to one side of the midline of the disc space, or a first implant to be positioned spaced laterally to one side of the midline of the disc space and a second implant to be positioned spaced laterally to the opposite side of the midline of the disc space.

(10) In one embodiment, the distal beveled end surface connects to the lateral walls of the endplates opposite the position of the distal end hook portion via a curved portion. The curved portion allows the implant to achieve increased lateral contact with bone close to the anterior apophyseal rim.

(11) In some embodiments, various other implants, systems and methods are disclosed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure is further informed by the specific description accompanied by the following drawings, in which:

(2) FIG. 1 is a perspective view of one embodiment of an expandable spinal implant system in a closed configuration in accordance with the principles of the present disclosure;

(3) FIG. 2 is a perspective view one embodiment of an expandable spinal implant system in an open configuration in accordance with the principles of the present disclosure;

(4) FIG. 3 is a perspective view of the components shown in FIG. 1 but with one endplate removed to show inner structures of a closed expandable spinal implant system in accordance with the principles of the present disclosure;

(5) FIG. 3A is a perspective view of an endplate component in accordance with the principles of the present disclosure;

(6) FIG. 4 is a perspective view of the components shown in FIG. 1 but with one endplate removed to show inner structures of an open expandable spinal implant system in accordance with the principles of the present disclosure;

(7) FIG. 5 is a perspective view of one embodiment of an expandable spinal implant system in a closed configuration in accordance with the principles of the present disclosure;

(8) FIG. 6 is a perspective view one embodiment of an expandable spinal implant system in an open configuration in accordance with the principles of the present disclosure;

(9) FIG. 7 is a perspective view of the components shown in FIG. 5 but with one endplate removed to show inner structures of a closed expandable spinal implant system in accordance with the principles of the present disclosure;

(10) FIG. 8 is a perspective view of the components shown in FIG. 6 but with one endplate removed to show inner structures of an open expandable spinal implant system in accordance with the principles of the present disclosure;

(11) FIG. 9 is a perspective view of the components of an expandable spinal implant system including an insertion instrument engaged with an expandable spinal implant in accordance with the principles of the present disclosure;

(12) FIG. 10 is a perspective view of the components shown in FIG. 9 also showing a driver shaft extended through the cannula and in engagement with the plug;

(13) FIG. 11 is a perspective view of the components shown in FIG. 9 also showing a driver shaft extended through the cannula and in engagement with the plug to expand the endplates relative to the frame;

(14) FIG. 12 is a perspective view of the components shown in FIG. 9 also showing the driver shaft

removed from the cannula;

(15) FIG. 13 is a top view of one embodiment of an expandable spinal implant system as used in a PLIF surgical procedure in accordance with the principles of the present disclosure;

(16) FIG. 14 is a perspective view of the components shown in FIG. 13 as used in a PLIF surgical procedure in accordance with the principles of the present disclosure;

(17) FIG. 15 is a top view of one embodiment of an expandable spinal implant system as used in a TLIF surgical procedure in accordance with the principles of the present disclosure;

(18) FIG. 16 is a perspective view of the components shown in FIG. 15 as used in a TLIF surgical procedure in accordance with the principles of the present disclosure;

(19) FIG. 17 is a perspective view of the components shown in FIG. 15 as used in a TLIF surgical procedure in accordance with the principles of the present disclosure;

(20) FIG. 18 is a perspective view of one embodiment of an expandable spinal implant system with a single movable endplate and wherein the frame may be substantially integral with at least one endplate;

(21) FIG. 19 is a perspective view of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion;

(22) FIG. 20 is a top view of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion;

(23) FIGS. 21 and 22 are perspective views depicting an initial insertion path into a disc space of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion;

(24) FIG. 23 is a perspective view depicting an insertion path into a disc space of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion hooking around a vertebral foramen of a vertebral body;

(25) FIG. 24 is a perspective view depicting a continued insertion path into a disc space toward an anterior apophyseal rim of a vertebral body of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion;

(26) FIG. 25 is a perspective view depicting one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion, positioned between upper and lower vertebral bodies, expanding toward an expanded position;

(27) FIG. 26 is a perspective view depicting one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion positioned on a lower vertebral body, expanding toward an expanded position;

(28) FIG. 27 is a perspective view of one embodiment of an expandable spinal implant, in a collapsed position, having an anterior projection and an anterior tip defining a distal end hook portion, including backout-prevention portions defined on the upper endplate;

(29) FIG. 28 is an upper view of one embodiment of an expandable spinal implant, in a collapsed position, having an anterior projection and an anterior tip defining a distal end hook portion, including backout-prevention portions defined on the upper endplate;

(30) FIG. 29 is a perspective view of one embodiment of an expandable spinal implant, in an expanded position, having an anterior projection and an anterior tip defining a distal end hook portion, including backout-prevention portions defined on the upper endplate;

(31) FIG. 30 is a perspective view of one embodiment of an expandable spinal implant having an anterior projection, an anterior tip defining a distal end hook portion, and an arcuate portion at the distal end opposite the distal end hook portion;

(32) FIG. 31 is an upper view of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion;

(33) FIG. 32 is a perspective view is a perspective view of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion;

(34) FIG. 33 is an upper view of one embodiment of an expandable spinal implant having an

anterior projection and an anterior tip defining a distal end hook portion inserted into a disc space with the distal end hook portion proximate the anterior apophyseal rim, and a proximal end of the implant proximate the vertebral foramen;

(35) FIG. **34** is an upper view of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion being inserted into a disc space;

(36) FIG. **35** is an upper view of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion being inserted into a disc space, with the distal end hook portion hooking around the vertebral foramen, thereby avoiding contact with at least the spinal cord;

(37) FIG. **36** is an upper view of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end being inserted into a disc space, with the distal end hook portion rotating to a TLIF/transverse pathway;

(38) FIG. **37** is an upper view of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion inserted into the disc space to a position proximate the apophyseal rim, and crossing over the midline of the disc space;

(39) FIG. **38** is an upper view of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion inserted into the disc space and spaced laterally away from the midline of the disc space;

(40) FIG. **39** is an upper view of two expandable spinal implants inserted into a disc space, each implant having an anterior projection and an anterior tip defining a distal end hook portion inserted into the disc space and spaced laterally away from the midline of the disc space;

(41) FIG. **40** is a perspective view of one embodiment of an expandable spinal implant having an anterior projection and an anterior tip defining a distal end hook portion in an orientation that is reversed from the orientation depicted in previous embodiments; and

(42) FIG. **41** is an upper view of the expandable spinal implant of FIG. **40** being inserted into the disc space toward the apophyseal rim.

DETAILED DESCRIPTION

(43) The exemplary embodiments of the surgical system and related methods of use disclosed are discussed in terms of medical devices for the treatment of musculoskeletal disorders and more particularly, in terms of an expandable surgical implant system that may include an expandable spinal implant, an insertion instrument and/or a method for treating a spine.

(44) In some embodiments, the present system includes an expandable spinal implant system suitable for insertion from a direct posterior (sometimes referred to as PLIF procedures) in pairs or singularly and then expandable at a distal end in order to impart and/or augment a lordotic curve of the spine. In some embodiments shown herein, the expandable spinal implant system may also be configured for use in oblique, postero-lateral procedures and/or transforaminal lumbar interbody fusions (sometimes referred to as TLIF procedures). Additionally, the frame disclosed in various embodiments may be configured to place a movable plug of the spinal implant in a substantially distal position within the spinal implant so as to clear a proximal volume within the implant for packing with bone-growth promoting materials after the implant has been inserted and/or expanded using the various techniques described herein. The frame and other various spinal implant components may also be configured with one or more sidewalls and/or openings to direct bone-growth promoting material to a selected area of an intervertebral or interbody space after the insertion and/or deployment of the spinal implant. In some embodiments, the spinal implant system may also be provided with a tapered distal tip (as viewed from a superior or top surface) such that the implant is shaped for insertion from an oblique approach and placement at a diagonal across an intervertebral or interbody space.

(45) In some embodiments, the spinal implant system may also be employed to restore and/or impart sagittal balance to a patient by increasing and/or restoring an appropriate lordotic angle

between vertebral bodies at a selected level where the spinal implant is implanted and expanded. In some embodiments, a pair of such spinal implants may be employed from bilateral PLIF approaches and expanded to differing heights to impart and/or restore both a lordotic angle as well as align the spine in the coronal plane (so as to treat a scoliotic curvature, for example). In some embodiments, a single such spinal implant may be employed from a postero-lateral TLIF approach and expanded to differing heights to impart and/or restore both a lordotic angle as well as align the spine in the coronal plane (so as to treat a scoliotic curvature, for example). In the various embodiments described, the spinal implant system may be useful in a variety of complex spinal procedures for treating spinal conditions beyond one-level fusions. Furthermore, the spinal implant system described in the enclosed embodiments may also be used as a fusion device with an expandable height for tailoring the implant to a particular interbody disc space to restore the spacing between adjacent vertebral bodies and facilitate spinal fusion between the adjacent vertebral bodies.

(46) In some embodiments, and as mentioned above, the present disclosure may be employed to treat spinal disorders such as, for example, degenerative disc disease, disc herniation, osteoporosis, spondylolisthesis, stenosis, scoliosis and other curvature abnormalities, kyphosis, tumor and fractures. In some embodiments, the present disclosure may be employed with other osteal and bone related applications, including those associated with diagnostics and therapeutics. In some embodiments, the disclosed spinal implant system may be alternatively employed in a surgical treatment with a patient in a prone or supine position, and/or employ various surgical approaches to the spine, including anterior, posterior, posterior mid-line, direct lateral, postero-lateral oblique, and/or antero lateral oblique approaches, and in other body regions. The present disclosure may also be alternatively employed with procedures for treating the lumbar, cervical, thoracic, sacral and pelvic regions of a spinal column. The spinal implant system of the present disclosure may also be used on animals, bone models and other non-living substrates, such as, for example, in training, testing and demonstration.

(47) The present disclosure may be understood more readily by reference to the following detailed description of the embodiments taken in connection with the accompanying drawing figures, which form a part of this disclosure. It is to be understood that this application is not limited to the specific devices, methods, conditions or parameters described and/or shown herein, and that the terminology used herein is for the purpose of describing particular embodiments by way of example only and is not intended to be limiting. In some embodiments, as used in the specification and including the appended claims, the singular forms “a,” “an,” and “the” include the plural, and reference to a particular numerical value includes at least that particular value, unless the context clearly dictates otherwise. Ranges may be expressed herein as from “about” or “approximately” one particular value and/or to “about” or “approximately” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. It is also understood that all spatial references, such as, for example, horizontal, vertical, top, upper, lower, bottom, left and right, are for illustrative purposes only and can be varied within the scope of the disclosure. For example, the references “upper” and “lower” are relative and used only in the context to the other, and are not necessarily “superior” and “inferior”.

(48) As used in the specification and including the appended claims, “treating” or “treatment” of a disease or condition refers to performing a procedure that may include administering one or more drugs, biologics, bone grafts (including allograft, autograft, xenograft, for example) or bone-growth promoting materials to a patient (human, normal or otherwise or other mammal), employing implantable devices, and/or employing instruments that treat the disease, such as, for example, micro-discectomy instruments used to remove portions bulging or herniated discs and/or bone spurs, in an effort to alleviate signs or symptoms of the disease or condition. Alleviation can occur

prior to signs or symptoms of the disease or condition appearing, as well as after their appearance. Thus, treating or treatment includes preventing or prevention of disease or undesirable condition (e.g., preventing the disease from occurring in a patient, who may be predisposed to the disease but has not yet been diagnosed as having it). In addition, treating or treatment does not require complete alleviation of signs or symptoms, does not require a cure, and specifically includes procedures that have only a marginal effect on the patient. Treatment can include inhibiting the disease, e.g., arresting its development, or relieving the disease, e.g., causing regression of the disease. For example, treatment can include reducing acute or chronic inflammation; alleviating pain and mitigating and inducing re-growth of new ligament, bone and other tissues; as an adjunct in surgery; and/or any repair procedure. Also, as used in the specification and including the appended claims, the term “tissue” includes soft tissue, ligaments, tendons, cartilage and/or bone unless specifically referred to otherwise. The term “bone growth promoting material” as used herein may include, but is not limited to: bone graft (autograft, allograft, xenograft) in a variety of forms and compositions (including but not limited to morselized bone graft); osteoinductive material such as bone morphogenetic proteins (BMP) (including but not limited to INFUSE® available from Medtronic plc) and alternative small molecule osteoinductive substances; osteoconductive materials such as demineralized bone matrix (DBM) in a variety of forms and compositions (putty, chips, bagged (including but not limited to the GRAFTON® family of products available from Medtronic plc)); collagen sponge; bone putty; ceramic-based void fillers; ceramic powders; and/or other substances suitable for inducing, conducting or facilitating bone growth and/or bony fusion of existing bony structures. Such bone growth promoting materials (denoted “BG” in some Figures herein) may be provided in a variety of solids, putties, liquids, colloids, solutions, or other preparations suitable for being packed or placed into or around the various implant **10**, **20** embodiments described herein.

(49) The following discussion includes a description of a surgical system including one or more spinal implants, related components and methods of employing the surgical system in accordance with the principles of the present disclosure. Various alternate embodiments are disclosed and individual components of each embodiment may be used with other embodiments. Reference is made in detail to the exemplary embodiments of the present disclosure, which are illustrated in the accompanying figures. Turning to FIGS. **1-12**, there are illustrated components of a surgical system, such as, for example, an expandable spinal implant **10**, **20** and associated system including an insertion instrument **30**.

(50) The components of expandable spinal implant system **10**, **20**, **30** can be fabricated from biologically acceptable materials suitable for medical applications, including metals, synthetic polymers, ceramics and bone material and/or their composites. For example, the components of expandable spinal implant system (including, but not limited to implant **10**, implant **20**, insertion instrument **30**), individually or collectively, can be fabricated from materials such as stainless steel alloys, commercially pure titanium, titanium alloys, Grade 5 titanium, super-elastic titanium alloys, cobalt-chrome alloys, stainless steel alloys, superelastic metallic alloys (e.g., Nitinol, super elastoplastic metals, such as GUM METAL®), ceramics and composites thereof such as calcium phosphate (e.g., SKELITE™), thermoplastics such as polyaryletherketone (PAEK) including polyetheretherketone (PEEK), polyetherketoneketone (PEKK) and polyetherketone (PEK), carbon-PEEK composites, PEEK-BaSO₄ polymeric rubbers, polyethylene terephthalate (PET), fabric, silicone, polyurethane, silicone-polyurethane copolymers, polymeric rubbers, polyolefin rubbers, hydrogels, semi-rigid and rigid materials, elastomers, rubbers, thermoplastic elastomers, thermoset elastomers, elastomeric composites, rigid polymers including polyphenylene, polyamide, polyimide, polyetherimide, polyethylene, epoxy, bone material including autograft, allograft, xenograft or transgenic cortical and/or corticocancellous bone, and tissue growth or differentiation factors, partially resorbable materials, such as, for example, composites of metals and calcium-based ceramics, composites of PEEK and calcium based ceramics, composites of PEEK with

resorbable polymers, totally resorbable materials, such as, for example, calcium based ceramics such as calcium phosphate, tri-calcium phosphate (TCP), hydroxyapatite (HA)-TCP, calcium sulfate, or other resorbable polymers such as polyaetide, polyglycolide, polytyrosine carbonate, polycaroplaetohe and their combinations.

(51) Various components of spinal implant system **10** may be formed or constructed material composites, including the above materials, to achieve various desired characteristics such as strength, rigidity, elasticity, compliance, biomechanical performance, durability and radiolucency or imaging preference. The components of expandable spinal implant system **10, 20, 30**, individually or collectively, may also be fabricated from a heterogeneous material such as a combination of two or more of the above-described materials. The components of expandable spinal implant system **10, 20, 30** may be monolithically formed, integrally connected or include fastening elements and/or instruments, as described herein. For example, in some embodiments expandable spinal implant system **10, 20, 30** may comprise expandable spinal implants **10, 20** comprising PEEK and/or titanium structures with radiolucent markers (such as tantalum pins and/or spikes) selectively placed in the implant to provide a surgeon with placement and/or sizing information when the expandable spinal implant **10, 20** is placed in the spine. The components of expandable spinal implant system **10, 20, 30** may be formed using a variety of subtractive and additive manufacturing techniques, including, but not limited to machining, milling, extruding, molding, 3D-printing, sintering, coating, vapor deposition, and laser/beam melting. Furthermore, various components of the expandable spinal implant system **10, 20, 30** may be coated or treated with a variety of additives or coatings to improve biocompatibility, bone growth promotion or other features. For example, the endplates **140, 150, 240, 250** may be selectively coated with bone growth promoting or bone ongrowth promoting surface treatments that may include, but are not limited to: titanium coatings (solid, porous or textured), hydroxyapatite coatings, or titanium plates (solid, porous or textured).

(52) Expandable spinal implant system **10, 20, 30** may be employed, for example, with a minimally invasive procedure, including percutaneous techniques, mini-open and open surgical techniques to deliver and introduce instrumentation and/or one or more spinal implants at a surgical site within a body of a patient, for example, a section of a spine. In some embodiments, expandable spinal implant system **10, 20, 30** may be employed with surgical procedures, as described herein, and/or, for example, corpectomy, discectomy, fusion and/or fixation treatments that employ spinal implants to restore the mechanical support function of vertebrae. In some embodiments, expandable spinal implant system **10, 20, 30** may be employed with surgical approaches, including but not limited to: posterior lumbar interbody fusion (PLIF), oblique lumbar interbody fusion, transforaminal lumbar interbody fusion (TLIF), various types of anterior fusion procedures, and any fusion procedure in any portion of the spinal column (sacral, lumbar, thoracic, and cervical, for example). Exemplary use of the expandable spinal implant system **10, 20, 30** in PLIF and TLIF techniques is shown generally in FIGS. **13-17**.

(53) As shown generally in FIGS. **1-8**, two exemplary embodiments of an expandable spinal implant **10, 20** are shown (implant **10** is highlighted in exemplary FIGS. **1-4** and implant **20** is highlighted in exemplary FIGS. **5-8**). Referring to FIGS. **1** and **2**, expandable spinal implant **10** may comprise a frame **100** comprising a proximal wall **110** and a distal wall **120**. The frame **100** may provide a mechanism for placing an expansion mechanism distally in the implant **10** such that, once expanded, the implant **10** provides ample room nearer the proximal end of the implant (such as at least partially within the frame **100**, for example) for the post-packing of bone growth promoting materials. For example, the proximal wall **110** of the frame **100** may define a proximal aperture **111** which may be suitable for receiving at least part of an insertion instrument **30** through which bone growth promoting material may be introduced into a proximal portion of the implant **10**. Furthermore, the distal wall **120** of the frame may define a distal aperture **121** (see FIG. **2**, for example) that is adapted to receive a plug **130**. As described further herein, the plug **130** may be

movably disposed in the distal aperture **121** of the frame.

(54) A plunger, syringe, tamp, funnel, pistol grip or hydraulic means may optionally be used to advance the material down the insertion instrument **30** and into the implant **10** and/or disc space. Another alternative may include passing a tube down the insertion instrument **30**. The tube in one preferred embodiment may be flexible and made of plastic or rubber. The tube may be prefilled with graft or other material and in one embodiment has a syringe attached at the end of the tube or alternatively includes a pistol grip, funnel or other means of advancing graft or other material down the tube. The syringe may be prefilled with material. The tube may be tapered at the distal end to facilitate interfacing the implant **10**. The tube and or the insertion instrument **30** may be flared out at the proximal end to act as a funnel or to facilitate receiving a funnel, syringe, pistol grip, or other instrument for providing and delivering the material. The graft also may be loaded into the disc space, where disc material has been removed, prior to the insertion of the implant **10**, by a tube such as disclosed in U.S. Pat. No. 8,092,464 to McKay, incorporated by reference herein in its entirety. The graft loading process described here and below is not necessarily limited to just use with the disclosed expandable implant but rather could be used in any fusion procedure or with the use of any intervertebral implant (lateral, oblique, ALIF, PLIF, etc.).

(55) The expandable spinal implant **10** may further comprise a first endplate **140** operably engaged with the frame **100** and configured to expand outward from the frame **100** when the plug **130** is moved in a distal direction D (See FIGS. 3-4). Furthermore, in some embodiments, the expandable spinal implant **10** may comprise opposing first and second endplates **140**, **150** as shown generally in FIGS. 1 and 2. In some such embodiments of the expandable spinal implant **10**, the second endplate **150** may be operably engaged with the frame **100** and configured to expand outward from the frame **100** when the plug **130** is moved in a distal direction D. Furthermore, as shown in FIG. 1, the second endplate **150** may be disposed about the frame **100** and opposing the first endplate **140**, wherein the first endplate **140** and the second endplate **150** extend from a proximal end of the implant **10** to a distal end of the implant **10** (along the length L of the implant **10**) and at least partially enclose the frame **100**. A similar structure is also shown in implant **20** of FIGS. 5-8, wherein endplates **240**, **250** cooperate to at least partially enclose the frame **200** (see FIG. 5, for example). The various endplates **140**, **150**, **240**, **250** may be provided with convex surfaces in multiple planes to conform to adjacent vertebral body endplates (see V1, V2 as shown in FIGS. 14 and 16). It should be understood that the surfaces of the various endplates **140**, **150**, **240**, **250** could also be constructed with a convexity in only one plane or without any convexities. Furthermore, the vertebral body V1, V2 contacting surfaces of endplates **140**, **150**, **240**, **250** may be provided with various anti-migration and/or osseointegration features including, but not limited to: ridges, teeth, pores, and coatings (including but not limited to porous titanium coatings such as those provided on Capstone PTC™ implants available from Medtronic plc).

(56) FIG. 18 shows an embodiment of an expandable spinal implant **10** comprising only a first endplate **140** operably engaged with the frame **100** and configured to expand outward from the frame **100** when the plug **130** is moved in a distal direction D (See FIGS. 3-4). In the embodiment of FIG. 18, the second endplate **150** may be integrally formed with the frame **100** and/or non-movable relative to the frame **100** such that as the plug **130** is moved distally, only the first endplate **140** (hinged to the frame **100** via pin **154**). In such embodiments, the distal head portion **135** may be modified to engage the movable first endplate **140** and the static second endplate **150**. For example, as shown generally in FIG. 3A, the movable first or second endplate **150** (and/or the complementary endplate **140**) may comprise a ramped surface **153** upon which ramped surface **136** of the distal head portion **135** may bear as the implant **10** is expanded. The ramp **136/153** mechanism may cooperate with a paired lateral post **137** and track **145** system (see FIG. 18) in order to optimize the opening and/or expansion of the implant **10**.

(57) Referring generally to FIGS. 1-4, the endplates **140**, **150** may be operably engaged with the frame **100** via a hinge mechanism located near or on the proximal wall **110** of the frame **100**. For

example, pins **154** may be provided that engage corresponding pin apertures **112** defined in the frame **100** such that the endplates are operably engaged with and/or hinged relative to the frame **100** such that the endplates **140**, **150** may be expandable relative to the frame **100** by virtue of the cooperation of the pins **154** and pin apertures **112** as the plug **130** is moved distally D relative to the frame **100** of the implant **10**. Similar hinge mechanisms are also shown relative to the embodiments of FIGS. **7** and **8** comprising pins **154** engaged with pin apertures **212** to connect frame **200** with endplates **240**, **250** in a hinged relationship. While multi-part mechanical hinges are shown in some of the pictured embodiments, it should be understood that other types of hinge and/or connection mechanisms may also be used to operably engage the frame **100** with the expandable endplates **140**, **150** of the implant. For example, in some embodiments, a “living hinge” may be utilized wherein the endplates **140**, **150** are at least partially integrally formed with the frame **100** at the hinge point but with cut-outs or flex points that allow the endplates **140**, **150** to rotate about the hinge connection. In summary, the frame **100** and endplates **140**, **150** may be operably engaged in a number of different ways including but not limited to: integral connections, separable connections, mechanically fixed connections using fastener or adhesives, releasable connections (including, but not limited to keyways and partially open hinges), and other connection types. In some embodiments, the frame **100** and endplates **140**, **150** may be integrally formed using additive manufacturing techniques such as 3D printing or sintering laser/beam melting, casting, extruding, or machined in an integral form using subtractive manufacturing techniques from one or more stock materials.

(58) In some embodiments, the frame **100** of the expandable spinal implant **10** further comprises at least one side wall **102** engaged with the proximal wall **110** and the distal wall **120**. As shown generally in FIG. **3**, the side wall **102** or walls **102**, **104** may be configured to space the proximal wall **110** and the distal wall **120** along a longitudinal axis (running substantially and/or nearly parallel to the length L) of the expandable spinal implant **10**. The side walls **102**, **104** may also be configured to contain bone growth promoting material in a proximal portion of the implant **10** that may be pre-packed or post-packed into the implant **10** via the proximal aperture **111**. The side walls **102**, **104** may cooperate with the proximal wall **110** and the distal wall **120** to create a four-sided frame **100** (that may define side apertures as shown in FIGS. **3-4**). In some such embodiments, the frame may define internal threads **103** configured to cooperate with an outer threaded surface **131** of the plug **130** when the plug **130** is positioned generally proximally relative to the distal wall **120** of the frame **100**.

(59) The frame **100** may be especially useful in some embodiments for placing the plug **130** in a substantially distal position relative to the overall length L of the implant **10** such that a distal portion of the implant (within a volume substantially encompassed by the frame **100**, for example) may be open and free to be filled (or “post-packed” with bone-growth promoting materials after the implant has been placed in a disc space between vertebral bodies (see, for example, the placement of implant **10**, between vertebral bodies V1 and V2, shown in FIGS. **14** and **16**). As described herein with respect to FIGS. **1**, **3**, **5** and **7**, the implant **10**, **20** may comprise or define a length L along a central longitudinal axis thereof, CL, extending from a proximal end **110** thereof to a distal end **144** thereof. In some such embodiments, the distal wall **120** of the frame may be disposed at least one third ($\frac{1}{3}$) of the length L (i.e. at a position spaced distally from the proximal end **110** by a distance W as shown generally in FIGS. **3** and **7**). In other embodiments the distal wall **120** of the frame may be disposed at other fractions of the length L (i.e. at a position spaced distally from the proximal end **110** by a distance W as shown generally in FIGS. **3** and **7**) including, but not limited to, at least $\frac{1}{10}$, $\frac{1}{8}$, $\frac{1}{5}$, $\frac{1}{4}$, $\frac{3}{4}$, $\frac{7}{8}$ and $\frac{9}{10}$. In other embodiments, the distal wall **120** of the frame may be disposed at a position spaced distally from the proximal end **110** by a distance W as shown generally in FIGS. **3** and **7** wherein the distance W ranges from 0 to 100 percent of the distance L, but in some instances distance W is at least 0.25 of the distance L to provide space in a proximal portion of the implant **10** for bone growth promoting material to be adequately post-packed into the

area defined at least in part by distance W when the plug **130** is moved distally. Therefore a proximal portion of the implant **10** (such as an internal volume defined at least in part by frame **100**) may be left substantially open and in fluid communication with the proximal aperture **111** of the frame **100** such that a bone growth promoting material may be placed through the proximal aperture **111** of the frame **100** after the plug **130** is moved in a distal direction D (see FIG. 3 showing the plug in an initial position, and FIG. 4 showing the plug moved distally to reveal a frame **100** volume left open and in fluid communication with the proximal aperture **111**).

(60) In other embodiments, as shown relative to the implant **20** in FIGS. 5-8, a single side wall **204** may replace the dual-wall embodiments of FIGS. 1-4 to space the distal wall **120** of the frame **100** from the proximal wall **110** of the frame. In some such embodiments with a single side wall **204**, the frame **200** may be substantially open on one side of the implant **10** to allow for post-insertion packing of bone growth promoting material via the open side of the frame **200**. The “open” or wall-less side of the frame **200** (which may be positioned generally opposite the side wall **204**) may also be used to direct and/or contain bone growth promoting material that may be introduced to the implant implantation site through the proximal aperture **211** of the frame **200** of the implant **20**. As with the “closed” embodiment having two side walls **102**, **104**, the single side wall **204** embodiments may also define internal threads **203** configured to cooperate with an outer threaded surface **231** of the plug **230** when the plug **230** is positioned generally proximally relative to the distal wall **220** of the frame **200**.

(61) In various embodiments, the plug **130**, **230** provided in the expandable spinal implant **10**, **20** may comprise a threaded outer surface **131** (see FIG. 1, for example), and the distal aperture **121** may comprise a complementary threaded inner surface operably engaged with the threaded outer surface **131** of the plug **130**. The threaded outer surface **131** of the plug may be disposed on a proximal end of the plug **130** such that the plug **130** moves distally D as shown in FIG. 4 when the plug **130** is rotated relative to the distal wall **120** of the frame **100**. In some embodiments, as shown generally in FIGS. 4 and 8, the frame **100** may comprise a sidewall **104** connecting the distal wall **120** and the proximal wall **110**, wherein the at sidewall **104** comprises a sidewall threaded surface **103** configured to be operably engaged with the threaded outer surface **131** of the plug **130** (especially when the plug is still positioned proximally relative to the frame **100**). An alternate embodiment of the sidewall threaded surface **203** is also shown in FIG. 8. Furthermore, the plug **130** may also comprise a distal head portion **135** configured to urge the endplate **140** away from the frame **100** with the plug **130** is moved in a distal direction D. The distal head portion **135** may be configured in some embodiments (as shown generally in FIGS. 1-4) with a separate structure having ramped surfaces **136** that may be configured to interface with complementary ramped surfaces on the endplates **140**, **150**. For example, as shown in FIG. 3A, the endplate **150** (and the complementary endplate **140**) may comprise ramped surface **153** upon which ramped surface **136** of the distal head portion **135** may bear as the implant **10** is expanded. The ramp **136/153** mechanism may cooperate with the lateral post **137** and track **155** system in order to optimize the opening and/or expansion of the implant **10**. For example, the ramp **136/153** mechanism may provide a leading expansion mechanism that is subsequently assisted by the lateral post **137** and track **155** system to expand the implant as the plug **130** is moved. Furthermore, the lateral post **137** and track **155** system may also render the expansion of the implant **10** reversible by pulling the endplates **140**, **150** inward towards the frame **100** along a relatively smooth ramped incline provided by the ramp **136/153** mechanism. Furthermore, the plug **130** may comprise separate connecting elements **132**, **133** such that the distal head portion **135** of the plug may be distally movable relative to the frame **100** without rotation while a proximal portion of the plug **130** (such as that portion defining the threaded outer surface **131**) is able to freely rotate in the distal aperture **121** of the distal wall **120** of the frame **100**.

(62) In other embodiments, as shown generally in FIGS. 5-8, the plug **230** may include a distal head portion **235** comprising a tapered cylinder. In some such embodiments, the distal head portion

235 may be configured to rotate with the plug **230** and/or move only distally **D** relative to the frame **100** as a proximal portion (defining the outer threaded surface **231**, for example) is rotated relative to the frame **100** to drive the plug **230** in the distal direction **D**. According to some such embodiments, the distal head portion **235** may be configured to cooperate with a contoured bearing surface **253** (comprising in some instances a ramp and/or frusto-conical concave surface) defined on an interior surface of the endplates **240**, **250**.

(63) The distal head portions **135**, **235** may be configured in various ways to provide a lead-in or gradual taper in order to allow for an easier interaction between the plug **130**, **230** and the endplates **140**, **150** or **240**, **250**. For example, as shown generally in the partially disassembled view of FIG. 3 (where the first endplate **140**, is removed), the distal head portion **135** comprises a ramp **136** or wedge suitable for urging a complementary ramped or contoured surface **153** on the inside of the endplates **140**, **150** (see FIG. 3A, showing an isolated view of one endplate **150** with an exemplary ramp **153** formed therein) so as to gradually move the endplate **140** away from the frame **100** as the plug **130** is advanced distally along the length **L** of the implant **10**. Similarly, in the embodiments shown in FIGS. 5-8, the distal head portion **235** may be tapered to provide a lead-in or frustoconical shape that may be optimized with a taper that allows for a mechanical advantage to be realized when urging the endplates **240**, **250** away from the frame **200**. The resulting open configuration of the implant **20** is shown, for example, in FIG. 6. Furthermore, it should be understood that a variety of ramp and/or taper configurations may be used to optimize the interaction of the plug **130**, **230** with the endplates **140**, **150** or **240**, **250**. Such configurations may include, but are not limited to: sequential ramps or tapered frustoconical surfaces with varying angles; shallow angle sequential ramps or tapered frustoconical surfaces leading into higher angle sequential ramps or tapered frustoconical surfaces (increasing the mechanical advantage once an initial expansion of the implant **10** has been achieved), as well as other opening mechanisms (such as the lateral post **137** and track **155** system shown generally in FIGS. 2-4 that may combine to assist the ramps **136** (and **153**, See FIG. 3A) in expanding the implant **20**).

(64) As shown in FIGS. 2-4, in some embodiments of the expandable spinal implant **10**, the distal head portion **135** may comprise a lateral post **137** extending from the distal head portion **135** of the plug **130** and configured for cooperating with a corresponding channel **145**, **155** defined in the endplates **140**, **150**. The channels may be angled or partially angled to provide additional mechanisms for assisting in the expansion of the implant **10** as the plug **130** is advanced distally along the length **L** of the implant **10**. Referring more particularly, to FIG. 2, the first endplate **140** may define at least one lateral channel **145** configured to receive the lateral post **137** such that when the plug **130** is moved in a distal direction along the length **L**, the lateral post **137** of the distal head portion **135** is moved in a first direction in the lateral channel **145** to expand the first endplate **140** outward from the frame **100**. The post **131** and channel **145** mechanism may also aid in making the implant **10** expansion substantially reversible such that when the plug **130** is moved in a proximal direction (i.e. towards the distal wall **110** of the frame **100**) the lateral post **137** of the distal head portion **135** is moved in a second direction in the lateral channel **145** to contract the first endplate **140** towards the frame **100** (which may result in the implant **10** returning to the closed or unexpanded configuration shown generally in FIG. 1). This reversible feature, combined with the threaded mechanism of the plug **130** renders the implant **10** capable of being incrementally expanded or contracted through a substantially infinitely adjustable range of motion (bounded only by the length of the plug **130** and the corresponding bearing surfaces (see **253**, FIG. 6, for example) defined by the endplates of the implant **10**)).

(65) In some embodiments, the expandable spinal implant system **10**, **20** may be configured to be operable with and/or inserted by an insertion instrument **30** (see generally FIG. 17 for example). In some such embodiments, as shown in FIG. 9, the expandable spinal implant **10** may comprise a frame **100** comprising a proximal wall **110** and a distal wall **120**. The proximal wall **110** may further define a proximal aperture **111** and the distal wall **110** may further define a distal aperture

121. As described herein, one or both of the proximal aperture **111** and the distal apertures **121** may be internally threaded to receive other threaded components. In some embodiments, the proximal wall **110** may be adapted to receive an insertion instrument **30** (or in some cases an inner cannula **320** of the insertion instrument **30** as shown in FIG. 9).

(66) As described herein, the expandable spinal implant **10** may also comprise a plug **130** movably disposed in the distal aperture **121**, wherein the plug **130** comprises an interface **134** adapted to be operably engaged by at least a portion of the insertion instrument **30** to move the plug **130**. For example, in some embodiments, the insertion instrument **30** may comprise a driver shaft **330** with a driver on a distal end thereof (such as a hexalobular driver tip). The distal end of the driver shaft **330** may be engaged with the interface **134** of the plug **130** to rotate the plug in the distal aperture **121** of the frame **100** in order to expand the implant **10**. As described herein, expansion of the implant **10** may be achieved by the moving the endplates **140**, **150** that are operably engaged by the frame **100** and configured to move relative to the frame **100** when the plug **130** is moved by the insertion instrument **30** (or the driver shaft **330** thereof).

(67) As shown generally in FIG. 17, the driver shaft **330** may be coaxially disposed inside an inner cannula **320** of the insertion instrument **30**. Furthermore, both the driver shaft **330** and the inner cannula **320** may be coaxially disposed inside a cannula **310** of the insertion instrument **30**. Each of the driver shaft **330**, inner cannula **320** and cannula **310** may further be provided with various manipulation components **330'**, **320'** and **310'** respectively, so that the various components of the insertion instrument **30** may be operated and/or selectively manipulated independent of one another to perform various functions relative to the implant **10** (as described further herein).

(68) As described herein and shown in the embodiments of FIGS. 3 and 7, the frame **100**, **200** may further comprise at least one side wall **104**, **204** engaged with the proximal wall **110** and the distal wall **120** of the frame **100**. The side wall **104**, **204** may be configured to space the proximal wall **110** and the distal wall **120** of the frame **100** along a longitudinal axis (extending parallel to the length L) of the implant **10**, **20**. In some embodiments, as shown in FIG. 3, the frame **100** comprises a pair of side walls **102**, **104** spaced laterally apart and engaged with the proximal wall **110** and the distal wall **120** of the frame **100** to form a substantially closed area adapted to receive and/or contain a bone growth promoting material that may be placed through the proximal aperture **111** of the frame **100**. In some embodiments, the cannula **310** or inner cannula **320** of the insertion instrument **30** may be configured to convey bone growth promoting material through the insertion instrument **30** and into the area defined by the frame **100** when the implant **10** is in the expanded position (see FIG. 2, for example, showing the plug **130** moved distally forward and out of the proximal area of the implant **10** defined by the frame **100**).

(69) In some embodiments the frame **100** may be substantially “closed” with sidewalls as shown generally in FIGS. 9-12. In other embodiments, the frame **100** may comprise a pair of sidewalls **102**, **104** with lateral apertures as shown generally in FIGS. 1-4. In other embodiments, as shown generally in FIGS. 5-8, the frame **200** may comprise a unilateral or single side wall **204** forming a frame **200** with one “open” lateral side. In some such embodiments as shown in FIG. 8, the frame **200** may be adapted to at least partially contain a bone growth promoting material BG that may be placed through the proximal aperture **211** of the frame **200** and/or direct the bone growth promoting material BG outside of the expandable spinal implant **20** in a lateral direction between the proximal wall **210** and the distal wall **220** of the frame **200**.

(70) FIGS. 9-12 show various configurations of an implant **10** embodiment in use with an insertion instrument **30** to form an expandable spinal implant system according to one embodiment. As shown generally in FIG. 9, the system may comprise an insertion instrument **30** comprising a cannula **310** (which may include an inner cannula **320** and an outer cannula **310** as described herein) and a driver shaft **330** (see FIG. 10 and FIG. 17) removably and rotatably disposed within the cannula **310**. The system may also further comprise an expandable spinal implant **10** configured to be operably engaged with the insertion instrument **30** using a variety of mechanisms. As

described herein, the implant **10** comprises a frame **100** comprising a proximal wall **110** and distal wall **120**, wherein the proximal wall **110** defines a proximal aperture **111** and the distal wall **120** defines a distal aperture. The proximal wall **110** may be configured to receive a distal end of the cannula **310** (or the middle cannula **320**) for manipulating the expandable spinal implant **10**. For example, as shown in FIG. **9**, the cannula **310** may comprise prongs **311** configured for insertion into complementary receptacles **114** defined by the proximal wall **110** of the frame **100**. In other embodiments, the prongs **311** may interact with tabs or slots defined by the endplates **140**, **150**. The prongs **311** may interact with the receptacles **114** to enable a surgeon to manipulate the implant **10** effectively as it is engaged with a distal end of the insertion instrument. Furthermore, in some embodiments, the inner cannula **320** may comprise a threaded tip **321** configured for operably engaging threaded inner surface of the proximal aperture **111** of the frame **100**. In some such embodiments, the prongs **311** of the outer cannula may serve as an effective counter-torque device (preventing rotation of the implant **10** relative to the insertion instrument **30**) as the inner cannula **320** is rotated to engage the proximal aperture **111** of the frame **100**. FIG. **17** shows the insertion instrument **30** in relation to the implant **10** including manipulation components **330'**, **320'** and **310'** of the insertion instrument. For example, handle **310'** of the outer cannula **310** may be used to stabilize and/or manipulate the implant **10** even as the knob **320'** of the inner cannula **320** is rotated within the outer cannula **310** such that the threaded tip **321** may be engaged with the proximal aperture **111** of the frame **100** without rotating the implant **100**.

(71) As described herein, the implant **10** may be configured for expansion by virtue of a plug **130** movably disposed in the distal aperture **120** of the frame **100**. In some embodiments, the plug comprises a threaded outer surface **131** configured to be engaged with a complementary inner threaded surface of the distal aperture **120**. In some embodiments, as shown in FIG. **9**, the plug **130** may comprise an interface **134** configured to be operably engaged by a distal end of a driver shaft **330** to move (by threaded rotation, for example) the plug **130** relative to the frame. The driver shaft **330** may be coaxially placed within the cannula **310** and/or the inner cannula **310** and rotatable therein using the driver proximal end **330'** of the driver shaft **330**. The driver proximal end **330'** may comprise a keyed or faceted surface configured for engagement with a quick-release handle (not shown) or a powered driver (not shown) for rotating the driver shaft **330**. Furthermore, the plug interface **134** may comprise a drive receptacle configured to cooperate with a distal end of the driver shaft. The drive connection between the driver shaft **330** and the plug interface **134** may comprise a variety of drive interfaces including but not limited to: multi-lobular drives; hexalobular drives; cross or Phillips head drives; straight or “flat head” drives; square or other polygonal drives; and/or combinations thereof.

(72) As described herein, the movement of the plug **130** facilitated by the driver shaft **310** within the cannula **310** (and, in some cases the inner cannula **320**) may further cause the movement of an endplate **140**, **150** operably engaged with the frame **100** of the implant **10** relative to the frame **100** when the plug **130** is moved by the insertion instrument **30**. Thus the insertion instrument **30** (or the driver shaft **330** and driver proximal end **330'**) may be used to expand the endplates **140**, **150** relative to the frame **100** in order to selectively expand the implant **10** and/or impart a lordotic movement in adjacent vertebral bodies **V1**, **V2** as shown generally in FIGS. **14** and **16**. The length of the driver shaft **330** may be adjusted to account for the distal placement of the distal wall **120** of the frame **100** relative to the length **L** of the implant **10**. For example, the driver shaft **330** may be provided with a length that substantially exceeds that of the cannula **310** and/or inner cannula **320** so that the driver proximal end **330'** remains accessible and engaged with a handle or powered driver even when the driver shaft **330** remains engaged with the plug **130** of the implant **10** when the implant is in the fully expanded condition (see FIGS. **14** and **16**). This feature may be important in situations where a surgeon wishes to reverse the expansion of the implant **10** as described further herein with respect to the post **131** and channel **145** mechanisms of particular implant **10** embodiments.

(73) According to various embodiments, the driver shaft **330** may also be configured to be removable from the cannula **310** (and/or the inner cannula (if employed)), such that after the plug **130** of the implant **10** has been moved distally relative to the frame **100**, a bone growth promoting material BG may be introduced into the frame **100** of the expandable spinal implant **10** through the cannula **310** (and/or through the concentric inner cannula **320**, when used). The bone growth promoting material BG may be tamped or urged through the cannula **310** or inner cannula **310** using the driver shaft **330** or other tamp and/or rod (not shown) sized for slidable insertion through the cannula **310** and/or inner cannula **310**. A funnel (not shown) or other attachment may also be inserted into a proximal end of the cannula **310** or inner cannula **320** (such as at the point near the proximal end or knob **320'** of inner cannula **320**, as shown in FIG. **17**) to facilitate the introduction of the bone growth promoting material BG into the cannula **310** and/or inner cannula **320**.

(74) FIGS. **9-12** depict exemplary procedural steps for the use of the implant system in one embodiment. For example, FIG. **9** shows an unexpanded implant **10** attached to insertion device **30** using the prongs **311** of the cannula **310** and the distal end **321** of inner cannula **320**. The plug **130** is shown engaged with the distal aperture of distal wall **120** of the frame and the plug interface **134** is visible. In FIG. **10**, the driver shaft **330** is shown extended through cannula **310** and inner cannula **320** and engaged with the plug interface **134**. Referring to FIG. **17**, the driver proximal end **330'** may be rotated at this step to drive the plug **130** forward to expand the endplates **140**, **150** relative to the frame **100**. FIG. **11** shows the result of the interaction of the driver shaft **330** with the plug **130** and the distal movement of the plug **130** relative to the distal wall **120** of the frame **100** to expand the endplates **140**, **150** relative to the frame **100** of the implant **10**. FIG. **12** shows the insertion device **30** still engaged with the implant **10** but with the driver shaft **330** removed from the cannula **310** and inner cannula **320**, leaving the cannulas open for the introduction of bone growth promoting material BG through the insertion instrument **30** and into a proximal portion of the implant **10** defined generally by the now-open interior of the frame **100**.

(75) Referring to exemplary FIGS. **13-16**, spinal implant system **10**, **30** can be employed with a surgical arthrodesis procedure, such as, for example, an interbody fusion for treatment of an applicable condition or injury of an affected section of a spinal column and adjacent areas within a body, such as, for example, intervertebral disc space between a vertebra **V1** and a vertebra **V2**. In some embodiments, spinal implant system **10**, **30** can include an intervertebral implant that can be inserted with intervertebral disc space to space apart articular joint surfaces, provide support and maximize stabilization of vertebrae **V1**, **V2**. In some embodiments, spinal implant system **10**, **30** may be employed with one or a plurality of vertebra.

(76) A medical practitioner obtains access to a surgical site including vertebrae **V1**, **V2** such as through incision and retraction of tissues. Spinal implant system **10**, **30** can be used in any existing surgical method or technique including open surgery, mini-open surgery, minimally invasive surgery and percutaneous surgical implantation, whereby vertebrae **V1**, **V2** are accessed through a mini-incision, retractor, tube or sleeve that provides a protected passageway to the area. In one embodiment, the components of spinal implant system **10**, **30** are delivered through a surgical pathway to the surgical site along a surgical approach into intervertebral disc space between vertebrae **V1**, **V2**. Various surgical approaches and pathways may be used. FIG. **13** shows an example of a typical posterior lumbar interbody fusion (PLIF) approach using the spinal implant system **10**, **30** wherein a pair of implants **10** may be delivered, expanded to impart or restore a lordotic curve (see generally FIG. **14**), and then post-packed with bone growth promoting material BG after the removal of the driver shaft **330** from the insertion instrument **30**. As shown in FIG. **15**, unilateral approaches such as a transforaminal lumbar interbody fusion (TLIF) approach may also be used to place the implant in a substantially oblique position relative to the vertebrae **V1**, **V2**. In such procedures the distal end **144** of the endplates **140**, **150** may be shaped so that the implant **10** fits within the intervertebral space defined by the extents of the vertebral body **V2** as shown in FIG. **15**. Furthermore, in oblique placement applications the implant **10** endplates **140**, **150** may also be

provided with complementary oblique contact surfaces shaped to better impart and/or restore a lordotic curve as the implant **10** is expanded as shown generally in FIG. **16**. Furthermore, the endplates **140**, **150** of the implant may be provided with a variety of ridges, teeth, coatings or other surface treatments suitable for interacting with and/or securing relative to the adjacent vertebrae **V1**, **V2**.

(77) As will be appreciated by one of skill in the art, a preparation instrument (not shown) may be employed to remove disc tissue, fluids, adjacent tissues and/or bone, and scrape and/or remove tissue from endplate surfaces of vertebra **V1** and/or endplate surface of vertebra **V2** in preparation for the procedures utilizing the system **10**, **30**. In some embodiments, the size of implant **10** is selected after trialing using trialing instruments (not shown) that may approximate the size and configuration of the system **10**, **30** (as shown in FIG. **17**, for example). In some embodiments, such trials may be fixed in size and/or be fitted with expansion mechanisms similar to the various implant **10**, **20** embodiments described herein. In some embodiments, implant **10** may be visualized by fluoroscopy and oriented before introduction into intervertebral disc space. Furthermore, the insertion instrument **30** and implant **10** may be fitted with fiducial markers to enable image guided surgical navigation to be used prior to and/or during a procedure.

(78) In some embodiments as shown generally in FIGS. **13** and **15**, implant **10** provides a footprint that improves stability and decreases the risk of subsidence into tissue. In some embodiments as shown generally in FIGS. **14** and **16**, implant **10** provides angular correction, height restoration between vertebral bodies, decompression, restoration of sagittal and/or coronal balance and/or resistance of subsidence into vertebral endplates. In some embodiments, implant **10** engages and spaces apart opposing endplate surfaces of vertebrae **V1**, **V2** and is secured within a vertebral space to stabilize and immobilize portions of vertebrae **V1**, **V2** in connection with bone growth for fusion and fixation of vertebrae **V1**, **V2**.

(79) Components of spinal implant system **10**, **30** including implant **10** can be delivered or implanted as a pre-assembled device or can be assembled in situ. Components of spinal implant system **10**, **30** including implant **10** may be expanded, contracted, completely or partially revised, removed or replaced in situ. In some embodiments, one or all of the components of spinal implant system **10**, **30** can be delivered to the surgical site via mechanical manipulation and/or a free hand technique.

(80) In one embodiment, spinal implant system **10**, **30** includes a plurality of implants **10** (see FIG. **13** for one example). In some embodiments, employing a plurality of implants **10** can optimize angular correction and/or height restoration between vertebrae **V1**, **V2**. The plurality of implants **10** can be oriented in a side by side engagement, spaced apart and/or staggered.

(81) In some embodiments, spinal implant system **10**, **30** includes an agent, including but not limited to the bone growth promoting materials **BG** described herein, which may be disposed, packed, coated or layered within, on or about the components and/or surfaces of spinal implant system **10**, **30**. In some embodiments, the agent may include bone growth promoting material to enhance fixation of implant **10** with bony structures. In some embodiments, the agent may include one or a plurality of therapeutic agents and/or pharmacological agents for release, including sustained release, to treat, for example, pain, inflammation and degeneration.

(82) In one embodiment, implants **10**, **20** may include fastening elements, which may include locking structure, configured for fixation with vertebrae **V1**, **V2** to secure joint surfaces and provide complementary stabilization and immobilization to a vertebral region. In some embodiments, locking structure may include fastening elements, such as, for example, rods, plates, clips, hooks, adhesives and/or flanges. In some embodiments, the components of spinal implant system **10**, **30** can be used with screws to enhance fixation. The components of spinal implant system **10** can be made of radiolucent materials such as polymers. Radiopaque markers may be included for identification under x-ray, fluoroscopy, CT or other imaging techniques. The insertion instrument **30** alone or with the tube for insertion therethrough described above may be radiolucent

and may optionally include markers added at the distal tip and/or along the length of one or both of insertion instrument **30** and the tube to permit them to be seen on fluoroscopy/x-ray while advancing into the patient. If the implant **10** includes radiolucent markers placed near the proximal end this may permit visualization of the proximity of the tip of the tube moving toward the proximal end of the implant **10**.

(83) In one embodiment of the invention, the endplates **140** and **150** include an anterior portion **400** extending to one side of the implant from one side only of respective lateral surfaces at the respective distal ends of the endplates **140** and **150**. The anterior portion **400** extends along an axis **1-1**, which extends transverse to the central longitudinal axis CL. Anterior portion **400** ends in an anterior tip **402**. As depicted in FIGS. **19-23** and **27-32**, when the implant is in the collapsed position, the anterior portion **400** and the anterior tip **402** define a distal end hooked portion **404**. When the implant is in the collapsed position, the distal end of the first endplate **140** combines with the distal end of the second endplate **150** to define a distal end beveled portion **406**. The distal end hooked portion **404** provides each of the endplates **140/150** with an increased surface area, enabling them to withstand greater loads during implant expansion, and to create lordosis.

(84) In one embodiment of the invention, at least the distal end hooked portion **404** is configured, as depicted in FIGS. **23** and **35**, upon insertion of the implant **10** into a disc space between an upper vertebral body and a lower vertebral body, to hook around the vertebral foramen VF, thereby avoiding interference with the neural elements, particularly the spinal cord, located in the vertebral foramen.

(85) In one embodiment of the invention, at least the distal end hooked portion **404** is further configured, as depicted in FIG. **36**, upon insertion of the implant into the disc space, to rotate from a laterally divergent pathway to a TLIF/transverse pathway.

(86) In one embodiment of the invention, as depicted in FIGS. **24-26**, the implant **10** is configured to be inserted into the disc space until the distal end hooked portion **404** and the distal end beveled portion **406** engage an entire portion of the anterior apophyseal rims of each of the upper and lower vertebral bodies. As depicted in FIG. **33**, the distal end hook portion **404** and the distal beveled end portion **406** engage the anterior apophyseal rim, while a proximal end corner **408** contacts the posterior rim.

(87) In one embodiment of the invention, as depicted in FIG. **29**, the distal end hooked portion **404** and the distal end beveled portion **406** continue to engage entirely along apophyseal rims of each of the upper and lower vertebral bodies following expansion of the implant.

(88) In one embodiment of the invention, as depicted in FIGS. **27-32**, an upper surface **410** of the frame **100** includes first parallel rows of teeth **412**. A lower surface **414** of the frame **100** includes second parallel rows of teeth **416**. As depicted in FIGS. **27-32**, at least an upper surface **418** of the first endplate **140** includes third rows of teeth **420**, defined parallel to the first rows of teeth **412**. Fourth rows of teeth **422** are defined at least on the lower surface **421** of second endplate **150** in an orientation parallel to the first rows of teeth **412**.

(89) In one embodiment of the invention, as depicted in FIGS. **27-32**, an alternative embodiment of third rows of teeth **420'** can be provided on the upper surface **418** of first endplate **140** in an orientation perpendicular to the first rows of teeth **412**. Likewise an alternative embodiment of fourth rows of teeth **422'** can be provided on lower surface **421** of the second endplate **150** perpendicular to the distal end beveled portion **406**. Perpendicular third rows of teeth **420'** and fourth rows of teeth **422'** provide improved purchase against the vertebral bone of the anterior apophyseal rims when the implant **10** is in the expanded position. In addition, when the implant is in the expanded position, first rows of teeth **412**, second rows of teeth **416**, third rows of teeth **420**, fourth rows of teeth **422**, and alternate third rows of teeth **420'** and fourth rows of teeth **422'** all are configured to prevent inadvertent backout of the implant **10** from the anterior apophyseal rims of the upper and lower vertebral bodies. Although FIGS. **27-32** depict the respective surfaces of the endplates fully covered by the rows of teeth **420**, **422**, **420'**, and **422'**, it is within the scope of the

invention for these rows of teeth to only partially cover the surfaces of the endplates.

(90) In one embodiment of the invention, as depicted in FIG. **31**, a width **W1** of the implant **10** at the distal end hooked portion **404** is greater than a width **W2** of the implant **10** at the frame proximal wall **110**, thereby providing increased structural strength at the distal end hooked portion **404**.

(91) In one embodiment, as depicted in FIG. **37**, the distal end hooked portion **404** is configured to be positioned in the disc space between the upper and lower vertebral bodies at a position crossing the midline **ML** of the disc space.

(92) In one embodiment of the invention, as depicted in FIG. **38**, the distal end hooked portion **404** is configured to be positioned in the disc space between the upper and lower vertebral bodies at a position spaced laterally to one side of the midline **ML** of the disc space.

(93) In one embodiment of the invention, as depicted in FIG. **39**, a pair of expandable implants **10** is configured to be positioned in the disc space between the upper and lower vertebral bodies, each respective implant **10** being located at a position spaced laterally to opposite sides of the midline **ML** of the disc space.

(94) In one embodiment of the invention, as depicted in FIG. **31** the endplates include opposing respective sidewalls **423/424**. The endplate sidewalls **423/424**, with the frame sidewalls **102/104** and the proximal wall **110** define an enclosed area to hold bone growth material inserted into the implant **10** via the proximal aperture **111**.

(95) In one embodiment of the invention, as depicted in FIG. **29**, the distal head **135** of the plug **130** abuts against a distal wedge portion **426**. As the distal wedge portion **426** is pushed toward the proximal end of the implant **10**, it pivots the first endplate **140** away from the second endplate **150**, expanding the implant **10** to the expanded position.

(96) In one embodiment of the invention, as depicted in FIGS. **31-39**, the distal end beveled portion **406** intersects with one of the side walls **423/424** of the frame **100** opposite the distal end hooked portion **404**, defining a distal end arcuate portion **428**. The distal end arcuate portion **428** and the distal end beveled portion **406** together reduce a profile of insertion into the disc space of the implant **10**.

(97) In one embodiment of the invention, as depicted in FIG. **33**, the distal end arcuate portion **428** allows the implant **10** to obtain greater lateral bony contact which is closer to the harder lateral vertebral rim, and to provide a greater medial grafting area.

(98) In one embodiment of the invention, as depicted in FIGS. **40** and **41**, the orientation of the distal end hooked portion **404** is reversed to extend from the opposite side of the implant **10**. In this embodiment, as depicted in FIG. **41**, the implant **10** contacts both the anterior rim and the posterior rim, providing improved stabilization of the adjacent vertebral bodies.

(99) In some embodiments, the use of microsurgical, minimally-invasive and image guided technologies may be employed to access, view and repair spinal deterioration or damage, with the aid of spinal implant system **10**, **30**. Upon completion of the procedure, the non-implanted components, surgical instruments and assemblies (such as insertion instrument **30**) of spinal implant system **10**, **30** may be removed and the incision is closed. In some embodiments, the various instruments (such as the insertion instrumentation disclosed generally herein in FIG. **9** and related figures) disclosed may be provided with fiducial markers or other elements suitable for use with surgical navigation systems (including, but not limited to the STEALTHSTATION®. Navigation system available from Medtronic plc), such that a surgeon may view a projected trajectory or insertion pathway of the implants **10**, **20** relative to a patient's anatomy in real time and/or in near-real time.

(100) It will be understood that the various independent components of the expandable spinal implants **10**, **20**, systems and insertion instruments **30** described herein may be combined in different ways according to various embodiments. As a non-limiting example, the notches **114** shown in FIGS. **5-8** with respect to implant **20** may also be added to a proximal end of the implant

10 shown in FIGS. **1-4** As a further non-limiting example, the dual apertures **241a**, **241b**, **251a**, **251b** shown in FIGS. **5-8** with respect to the endplates **240**, **250** of implant **20**, may also be added to the endplates **140**, **150** of the implant **10** shown in FIGS. **1-4**.

(101) It will be understood that various modifications may be made to the embodiments disclosed herein. Other embodiments of the invention will be apparent to those skilled in the art from consideration of the specification and practice of the invention disclosed herein. It is intended that the specification and examples be considered as exemplary only, with the true scope and spirit of the invention being indicated by the following claims.

Claims

1. An expandable spinal implant comprising: a frame comprising a frame proximal end, an opposite frame distal end, a distal wall at the frame distal end, and a distal aperture defined in the distal wall; a plug movably disposed in the distal aperture and configured for movement from a first position to a second position, the plug including a first and a second engagement portion; a first endplate portion pivotally engaged with the frame and configured to expand outward from the frame when the plug is moved from the first position to the second position, the first endplate portion including an upper surface, a first endplate portion proximal end, a first endplate portion distal end, the first endplate portion being configured to engage the first engagement portion of the plug; a first distal end portion extending outwardly from the first endplate portion including at least one side surface and a first tip portion, the at least one side surface of the first distal end portion extending from the first tip portion to the first endplate portion; a second endplate portion pivotally engaged with the frame and configured to expand outward from the frame when the plug is moved from the first position to the second position, the second endplate portion including a lower surface, a second endplate portion proximal end, a second endplate portion distal end, the second endplate portion being configured to engage the second engagement portion of the plug; a second distal end portion extending outwardly from the second endplate portion including at least one side surface and a second tip portion, the at least one side surface of the second distal end portion extending from the second tip portion to the second endplate portion; wherein movement of the plug from the first position to the second position, and corresponding interaction of the first engagement portion with the first endplate portion causes the first endplate portion and the first distal end portion to move away from the frame and interaction of the second engagement portion with the second endplate portion causes the second endplate portion and the second distal end portion to move away from the frame to move the implant from a collapsed position to an expanded position; and wherein the first distal end portion includes a mid-longitudinal axis that is transverse to a mid-longitudinal axis of the first endplate portion, and the second distal end portion includes a mid-longitudinal axis that is transverse to a mid-longitudinal axis of the second endplate portion.

2. The expandable spinal implant of claim 1, wherein when the implant is in the collapsed position, the first tip portion and the second tip portion each further define a distal end beveled portion.

3. The expandable spinal implant of claim 2, wherein the implant is configured to be inserted in a disc space between upper and lower vertebral bodies until the distal end beveled portions of the first tip portion and the second tip portion engage an anterior apophyseal rim of one of the upper and lower vertebral bodies.

4. The expandable spinal implant of claim 1, wherein the first endplate upper surface comprises at least one first row of teeth defined thereon, and the second endplate lower surface comprises at least one second row of teeth defined thereon.

5. The expandable spinal implant of claim 1, wherein the frame further comprises a proximal wall at the frame proximal end, and at least one frame side wall engaged with the frame proximal wall and the frame distal wall, the frame side wall configured to space the frame proximal wall and the frame distal wall along a longitudinal axis of the expandable spinal implant.

6. The expandable spinal implant of claim 5, wherein the plug comprises a threaded outer surface, and wherein the distal aperture comprises a threaded inner surface operably engaged with the threaded outer surface of the plug.

7. The expandable spinal implant of claim 1, wherein a head portion of the plug is configured to contact portions of the first endplate portion and the second endplate portion to aid movement of the first endplate portion and the second endplate portion away from the frame when the plug is moved from the first position to the second position.

8. The expandable spinal implant of claim 1, wherein one of the first endplate portion and the second endplate portion further comprise a first lateral side wall and a second lateral side wall spaced laterally apart from one another, the first lateral side wall extending in a first plane, the second lateral side wall extending in a second plane, and at least portions of the first distal end portion and the second distal end portion being located outside of an area between the first plane and the second plane.

9. An expandable spinal implant comprising: a frame comprising a frame proximal end and an opposite frame distal end; a plug movably attached to the frame and configured for movement from a first position to a second position, the plug including a first and a second engagement portion; a first endplate portion pivotally engaged with the frame and configured to expand outward from the frame when the plug is moved from the first position to the second position via engagement of the first engagement portion of the plug with the first endplate portion; a first distal end portion extending outwardly from the first endplate portion and including a first tip portion; a second endplate portion pivotally engaged with the frame and configured to expand outward from the frame when the plug is moved from the first position to the second position via engagement of the second engagement portion of the plug with the second endplate portion; a second distal end portion extending outwardly from the second endplate portion and including a second tip portion; wherein movement of the plug from the first position to the second position, and corresponding interaction of the first engagement portion with the first endplate portion and of the second engagement portion with the second endplate portion causes the first endplate portion and the first distal end portion to move away from the frame and causes the second endplate portion and the second distal end portion to move away from the frame to move the implant from a collapsed position to an expanded position; and wherein the first distal end portion includes a mid-longitudinal axis that is transverse to a mid-longitudinal axis of the first endplate portion, and the second distal end portion includes a mid-longitudinal axis that is transverse to a mid-longitudinal axis of the second endplate portion.

10. The expandable spinal implant of claim 9, wherein when the implant is in the collapsed position, the first tip portion and the second tip portion each further define a distal end beveled portion; and wherein the implant is configured to be inserted in a disc space between upper and lower vertebral bodies until the distal end beveled portions of the first tip portion and the second tip portion engage an anterior apophyseal rim of one of the upper and lower vertebral bodies.

11. The expandable spinal implant of claim 9, wherein the first endplate upper surface comprises at least one first row of teeth defined thereon, and the second endplate lower surface comprises at least one second row of teeth defined thereon.

12. The expandable spinal implant of claim 9, wherein the frame further comprises a proximal wall at the frame proximal end and a distal wall at the frame distal end, and at least one frame side wall engaged with the frame proximal wall and the frame distal wall, the frame side wall configured to space the frame proximal wall and the frame distal wall along a longitudinal axis of the expandable spinal implant.

13. The expandable spinal implant of claim 12, wherein the plug comprises a threaded outer surface, and wherein the frame comprises a distal aperture with a threaded inner surface operably engaged with the threaded outer surface of the plug.

14. The expandable spinal implant of claim 9, wherein a head portion of the plug is configured to

contact inner surfaces of the first endplate portion and the second endplate portion to aid movement of the first endplate portion and the second endplate portion away from the frame when the plug is moved from the first position to the second position.

15. The expandable spinal implant of claim 9, wherein at least one of the first endplate portion and the second endplate portion further comprise a first lateral side wall and a second lateral side wall spaced laterally apart from one another, the first lateral side wall extending in a first plane, the second lateral side wall extending in a second plane, and at least portions of the first distal end portion and the second distal end portion being located outside of an area between the first plane and the second plane.

16. An expandable spinal implant comprising: a frame comprising a frame proximal end and an opposite frame distal end; a plug being moveable from a first position to a second position relative to the frame, and including at least one engagement portion; a first endplate portion engaged with the frame, the first endplate portion including a first endplate portion proximal end and a first endplate portion distal end; a first distal end portion extending outwardly from the first endplate portion and including a first tip portion; a second endplate portion engaged with the frame, the second endplate portion including a second endplate portion proximal end and a second endplate portion distal end; a second distal end portion extending outwardly from the second endplate portion and including a second tip portion; wherein at least one of the first endplate portion and the second endplate portion is pivotally attached to the frame, and includes at least one complimentary engagement portion for engaging the at least one engagement portion of the plug; wherein movement of the plug from the first position to the second position, and corresponding interaction of the at least one engagement portion with the at least one complimentary engagement portions causes movement of the implant from a collapsed position to an expanded position; and wherein the first distal end portion includes a mid-longitudinal axis that is transverse to a mid-longitudinal axis of the first endplate portion, and the second distal end portion includes a mid-longitudinal axis that is transverse to a mid-longitudinal axis of the second endplate portion.

17. The expandable spinal implant of claim 16, wherein the first distal end portion and the second distal end portion are each formed as a hooked end portion.

18. The expandable spinal implant of claim 16, wherein one of the first endplate portion and the second endplate portion further comprise a first lateral side wall and a second lateral side wall spaced laterally apart from one another, the first lateral side wall extending in a first plane, the second lateral side wall extending in a second plane, and at least portions of the first distal end portion and the second distal end portion being located outside of an area between the first plane and the second plane.

19. The expandable spinal implant of claim 16, wherein the frame further comprises a proximal wall at the frame proximal end and a distal wall at the frame distal end, and at least one frame side wall engaged with the frame proximal wall and the frame distal wall, the frame side wall configured to space the frame proximal wall and the frame distal wall along a longitudinal axis of the expandable spinal implant.

20. The expandable spinal implant of claim 19, wherein the plug comprises a threaded outer surface, and wherein the frame comprises a distal aperture with a threaded inner surface operably engaged with the threaded outer surface of the plug.
